

The empirical recurrence for OEIS A283039: recovering a conjecture that is not written down

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Abstract

OEIS A283039 records “Empirical recurrence of order 72”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 867$ vertices and satisfies some monic recurrence of order at most S ; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 72, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 72 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A283039 is “Number of $n \times 5$ 0..1 arrays with no 1 equal to more than one of its horizontal and vertical neighbors, with the exception of exactly one element.”. It has offset 1 and begins

5, 256, 5065, 94368, 1604067, 26180826, 414085368, 6406597648, ...

The entry states:

Empirical recurrence of order 72 (see link above).

As of the “Last modified” line on the live entry (Sep 03 2025 15:15:40, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 867$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S = 867$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 1734$ exact terms of the model returns order 72 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{72} c_i a(n-i),$$

c_1	10	c_{19}	10319468286	c_{37}	1394782470238	c_{55}	69639020
c_2	139	c_{20}	−1425827262	c_{38}	−708774302385	c_{56}	28366393
c_3	28	c_{21}	−37491905214	c_{39}	239160689324	c_{57}	−25049128
c_4	−10098	c_{22}	85431628908	c_{40}	27542439130	c_{58}	−9850164
c_5	−80100	c_{23}	−153547784370	c_{41}	−94555657876	c_{59}	2324752
c_6	−238934	c_{24}	180436319884	c_{42}	85275471695	c_{60}	1828313
c_7	257032	c_{25}	16536524900	c_{43}	−58845719780	c_{61}	892366
c_8	4044245	c_{26}	−550827934241	c_{44}	19619455035	c_{62}	−561101
c_9	9027286	c_{27}	1509686930684	c_{45}	−3504452266	c_{63}	−66966
c_{10}	−15041923	c_{28}	−2967035839405	c_{46}	5435629293	c_{64}	−17910
c_{11}	−97737144	c_{29}	4684556095012	c_{47}	2131615572	c_{65}	36276
c_{12}	−47245474	c_{30}	−6224930302728	c_{48}	−8106379706	c_{66}	−2241
c_{13}	464214514	c_{31}	7246992385190	c_{49}	3135328592	c_{67}	−880
c_{14}	553969670	c_{32}	−7422737805420	c_{50}	145394122	c_{68}	−591
c_{15}	−1348440058	c_{33}	6647605725346	c_{51}	752743416	c_{69}	124
c_{16}	−1382900685	c_{34}	−5302269887597	c_{52}	−409433266	c_{70}	13
c_{17}	1948008838	c_{35}	3816253056600	c_{53}	−331000266	c_{71}	2
c_{18}	−2260393309	c_{36}	−2439108709693	c_{54}	100770479	c_{72}	−1

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 72, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $72 + 4$ terms removed returns order 72 as well, so $\deg q_{\min} = 72 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 19 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $72 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 72 that this sequence satisfies — and that family turns out to have exactly one member.

References

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